

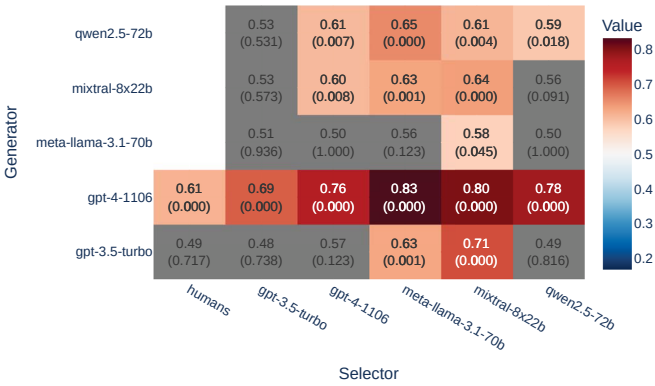


Fig. 3. Ratios of selectors preferring LLM-generated text over human-generated text for the paper dataset. Values represent the ratio of preferences for LLM-generated text, with *P*-values indicated in parentheses. Gray backgrounds denote results that are not statistically significant ($P > 0.05$), while zero *P*-values indicate highly significant results ($P < 0.0005$). The aggregated *P*-value, computed using Fisher's method (BH-corrected, $\alpha = 0.05$), confirms overall statistical significance despite variations in individual results: $P = 0.001$.

size of the human abstract because otherwise the LLM-generated abstracts would end up 2 to 3 $\times$ longer than the human ones and create an unfair comparison scenario. While the LLMs are not perfect at sticking precisely within the requested word count, the prompt modification had enough of an effect to get the human and LLM-generated abstracts to roughly equivalent lengths.

The results are displayed in Fig. 3. Consistent with the findings from the product experiment, abstracts generated by LLMs are generally preferred, though the effect is less pronounced.

3.3. Movie Plot Summaries Experiments. For this experiment, we analyzed whether LLMs prefer movies whose plot summaries were generated by LLMs over those written by humans.

To generate LLM-written summaries, we provided models with the movie title and year (All before 2013). The generation prompt used was the following:

1 Based on the title and year, write a plot summary of the following movie.

To examine whether LLMs exhibit bias toward AI-generated movie plot summaries, we presented them with pairs of summaries—one written by a human and one generated by an LLM. The models were then tasked with choosing which movie to recommend for purchase based solely on the plot summary.

1 The following are plot summaries of movies available for purchase. Which movie do you recommend choosing? Your client wants you to make a decision, so you have to choose only one of them, without additional context, even if the movie being described is identical in all of the options.

As shown in Fig. 4, we found that LLMs displayed a significant preference for movies with summaries generated by other LLMs, though the bias was weaker than in the product description experiment.

3.4. First-Item Bias. We define the first-item bias as the tendency of LLMs to select the first item they encounter when presented

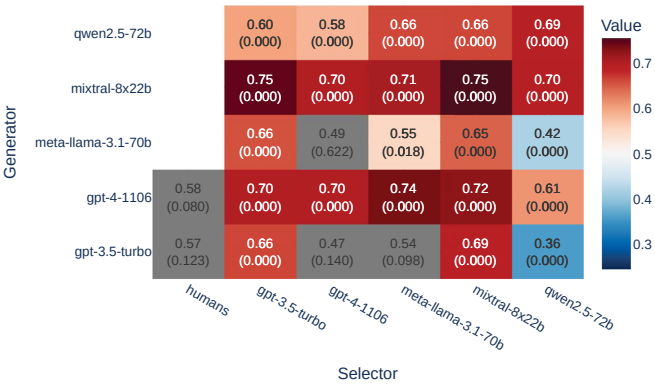


Fig. 4. Ratios of selectors preferring LLM-generated text over human-generated text for the movie dataset. Values represent the ratio of preferences for LLM-generated text, with *P*-values indicated in parentheses. Gray backgrounds denote results that are not statistically significant ($P > 0.05$), while zero *P*-values indicate highly significant results ($P < 0.0005$). Even with variation in individual results, the aggregated *P*-value remains significant: Fisher's method (BH-corrected, $\alpha = 0.05$) results in $P < 10^{-16}$.

with two choices (17, 18). We found that for some LLMs, this bias is moderately high when evaluated on descriptions generated by GPT-4. For example, GPT-3.5 exhibited a ratio of approximately 69% on the product dataset, while GPT-4 showed a ratio of around 73% on the movie dataset.

The first-item bias can be a problem, since if an LLM chooses the first option most of the time, it may obscure the true extent of its preference for LLM-generated content. For example, if the LLM selects the first item 80% of the time and in the remaining 20% of cases, the LLM selects LLM-generated content 90% of the time, the overall observed bias would appear to be 58%, while the true bias could be as high as 90% if the first-item effect were eliminated. Although first-item bias was present in almost all models, it was particularly strong in only some, as shown in the following sections and tables.[§]

To attempt to reduce the effects of a first-item bias, all comparisons between human and LLM-generated texts were done twice, with the order that the two texts were presented in the query swapped in between requests. However, our analysis does not explicitly account for the possibility that, even after averaging, first-item bias may still partially obscure the full extent of LLM-for-LLM bias. However, addressing this potential “masking” effect could lead to a stronger, hypothesis-affirming interpretation of our results for some models. Future work could explore methods to better isolate and quantify the potential masking effect of first-item bias.

3.4.1. Products. Table 1 presents the results of the first-item bias for the products experiments. Most notably, the Llama-3-70b-chat-hf model showed a significant bias of 73.18% compared to other models.

3.4.2. Paper abstracts. Table 2 presents the results of the first-item bias for the paper experiment. There, the models tend to be more in balance compared to the product experiments. Llama-3-70b-chat-hf tends to exhibit the highest first-item bias at 58.06%. Interestingly, GPT-4 and Qwen2.5-72B-Instruct-Turbo seem to prefer the second option more in this case.

[§]SI Appendix includes additional results for first-item bias across all used models as both generators and selectors.

Table 1. First-item bias of LLMs in product details recommendations

Model	Total	# Invalids	First option bias (%)
Qwen2.5-72B-Instruct-Turbo	220	0	56.81
Llama-3.1-70B-Instruct-Turbo	220	0	73.18
Mixtral-8x22B-Instruct-v0.1	220	11	53.59
GPT-3.5-turbo	220	44	46.59
GPT-4-1106-preview	220	3	56.68

3.4.3. Movie plot summaries. Table 3 presents the results of the first-item bias for the movie experiment.
For the movie plot summaries, GPT-4 exhibited the highest first-option bias at 73.46%.

3.5. Preferences of Humans. To complement our studies on LLM bias, we conducted an initial experiment to gauge human preferences in similar decision contexts (see Figs. 1–4 for results). It is important to note that these preferences were collected by research assistants rather than actual users, although we recruited assistants of diverse backgrounds,[§] using an online job board. This study serves as a preliminary investigation with a small sample size (13 in total; 6 for each dataset) and a best-effort human baseline, and the findings are not definitive.[#] We used the product descriptions, scientific paper abstracts, and movie plot summaries generated by both GPT-3.5 and GPT-4, and presented them to a group of human evaluators. These participants were asked to choose the item they preferred without knowing whether the description of the item was written by a human or an LLM. They also had the option to state that they had no preference between the two texts presented.

Participants were presented with pairs of descriptions: one generated by an LLM and one written by a human. Each participant evaluated a randomized set of pairs to mitigate any first-item bias.

The key metric we analyze is the ratio of human preference PNto LLM preference for LLM-pitched items (Tables 4 and 5). If both LLMs and humans prefer LLM-pitched items at equal rates, then we cannot rule out the possibility that LLMs’ preference for LLM-pitched items is driven by valid quality-signals. However, if LLMs show a significantly stronger preference for LLM-pitched items than do humans, LLMs’ preference for LLM-pitched items plausibly indicates a systematic LLM-for-LLM bias.

Except for a slight reversal in the movie domain with GPT-3.5,^{||} the results show that LLMs prefer LLM-pitched items at a higher rate than human evaluators. This discrepancy supports our design-driven hypothesis that the strong preference observed in LLMs toward LLM communications is driven by model-specific

Table 2. First-item bias of LLMs in academic paper recommendations

Model	Total	# Invalids	First option bias (%)
Qwen2.5-72B-Instruct-Turbo	186	2	40.22
Llama-3.1-70B-Instruct-Turbo	186	0	58.06
Mixtral-8x22B-Instruct-v0.1	186	1	53.51
GPT-3.5-turbo-1106	186	23	54.60
GPT-4-1106-preview	186	11	46.86

[§]SI Appendix includes demographic details about the human participants.
[#]SI Appendix also presents a leave-one-out analysis, which shows that, for the most part, removing any single participant does not substantially alter human preference patterns for LLM-pitched items.
^{||}One possible explanation is that two research assistants noted some human-written plot summaries were poorly crafted, which may have served as a negative quality signal or influenced their judgments affectively.

Table 3. First-item bias of LLMs in movie recommendations

Model	Total	# Invalids	First option bias (%)
Qwen2.5-72B-Instruct-Turbo	500	0	54.20
Llama-3.1-70B-Instruct-Turbo	500	2	67.67
Mixtral-8x22B-Instruct-v0.1	500	73	69.79
GPT-3.5-turbo	500	22	49.26
GPT-4-1106-preview	500	29	73.46

evaluation criteria rather than by quality-signal differences between human- and LLM-generated texts. Nonetheless, we encourage future work to conduct more extensive human studies to investigate this phenomenon in greater detail.

3.6. Experiment Implementation Details. Each comparison request to an LLM API involved two steps.

Step 1: First, the LLM was asked to choose between two texts, each assigned temporary integer IDs in the prompt, using the specific comparison prompt text as input, and receiving a free text response as output. An example prompt would be formatted like:

```
1 {comparison_prompt phrase}
2
3 ## {item type} {integer ID 1}:
4 {text 1}
5
6 ##$ {item type} {integer ID 2}:
7 {text 2}
8
9 ---
10 {optional addendum}
```

Step 2: Then a second independent query was sent to the LLM, with the prompt being the output of the first plus instructions to identify which text was selected and provide its integer ID in JSON format. The Interlab tool was used to aid the JSON request part (19). Resulting prompt format:

```
1 The following text is a snippet where the
  writer makes a choice between two items.
  Each {item type} should have an integer
  ID.
2 Which {item type} ID was chosen, if any?
3
4 ## Text snippet
5 {selection response from the first query}
6
7 ## Instructions to format the answer:
8 Write your answer to the prompt as a
  single JSON conforming to the
  following JSON schema:
9 “‘json
10 {‘type’: ‘object’, ‘properties’:
11 {‘answer’: {‘title’: ‘Answer’,
12 {‘description’: ‘The integer ID (one of
  the following:
13 {integer ID 1} or {integer ID 2}
14 of the item that was chosen, or None if
  no clear choice was made.’},
15 {‘type’: ‘integer’}}}}
16 “‘
17
18 The answer should contain exactly one
  markdown JSON code block delimited
19 by “‘‘‘json” and “‘‘‘”.
```

Table 4. Human-to LLM choice ratios with GPT-4 as text generator

Dataset	Human (%)	LLM (%)	Human/LLM
Product	36	89	0.40
Paper	61	78	0.78
Movie	58	70	0.83

4. Discussion

All three of our experiments show moderate-to-strong LLM preference for goods presented via LLM-authored promotional texts. Our experiments were designed to limit the possibility of a genuine quality-signals difference between the human and LLM “pitches,” such that strong preference for the LLM-pitched goods is suggestive of bias. To further test our interpretation of LLM-for-LLM preference as indicative of a bias, we repeated key parts of our experiments using human research assistants as selectors, operationalizing “LLM-for-LLM bias” as the difference between the frequency at which humans chose LLM pitches and the frequency at which LLMs chose LLM pitches. Although this was a small-scale, preliminary human test, the results on average show a moderate gap between our research assistants’ choices and those of the LLMs. Overall, we find that our original experiments together with the small-scale human test provide strong evidence for an LLM-for-LLM bias. Further research is needed to determine the scale of this bias as well as its ultimate sources. We discuss the assumptions and reasoning behind our interpretation of our results in more detail below.

Recall that our experiments instruct choice-makers to choose between goods presented via content-equivalent promotional texts: LLMs’ disposition to prefer objects promoted by LLM-authored texts (as observed in our three experiments) is tantamount to treating the textual correlates of LLM presentation as evidence of the superiority of the presented goods, or else equivalent to demonstrating a brute task-misaligned preference that tracks LLM author-identity. We defeasibly presume that treating the textual correlates of LLM presentation as evidenced of the pitched good’s superior quality is epistemically defective, making LLM-for-LLM preference defeasible evidence of an epistemically defective or task-misaligned disposition.

Since our experiment design cannot entirely eliminate the possibility of epistemically valid quality-signals bound to prose style,** we used a small-scale human test to “sanity-check” the presumption that LLM-for-LLM preference is not explained by genuine quality-signals. The underlying theory is that our human research assistants’ preferences give a plausible upper limit on genuine quality signals: in other words, that any plausible epistemically valid effect of LLM prose-style on preference

Table 5. Human-to LLM choice ratios with GPT-3.5 as text generator

Dataset	Human (%)	LLM (%)	Human/LLM
Product	29	66	0.44
Paper	49	60	0.82
Movie	57	56	1.01

**The potential sources for epistemically grounded preference in our experiments, which we believe to all be ultimately marginal, are diverse and differ from experiment to experiment: in the Papers experiment, for instance, eloquence in the abstract could potentially be treated as a signal of the competence of the paper’s author. Whereas in the Movies experiment differences in a summary’s emphasis or tone may “paint a different picture” of the summarized movie.

should also apply to our human research assistants and not only to LLM choice-makers. Additionally, we rely on the comparison with human preferences to begin distinguishing AI–AI implicit bias from more general pro-AI bias or “AI charisma” that affects humans’ judgment as well.

On average, we found that humans choose LLM-pitched goods less frequently than LLMs do. We did observe (somewhat contrary to our expectations) a weak human preference for LLM-pitched papers in the human test on the Papers Experiment^{††} and for LLM-pitched movies in the Movies experiment,^{‡‡} but we stress that this finding is consistent with a noticeable difference between humans’ and LLMs’ rates of favoring LLM-pitched papers.

Granting that our experiments show evidence of biased choice-dispositions in LLMs,^{§§} we now ask whether these choice-dispositions amount to (potential) “discrimination” in an ethically, socially, or economically relevant sense. While defining and testing discrimination in general is a complex and contested matter (20), our experiments are meant to diagnose (potential) discrimination in a specific narrow sense: by discrimination we mean an impactful choice-pattern that tracks the identity of candidates without thereby tracking other choice-relevant features. This usage aligns with an important operational definition of (one type of) discrimination in both the economics literature and machine learning literature, although the two literatures tend to conceptualize these choice-patterns in slightly different ways.^{¶¶} While there are many other, more conceptually complex forms of discrimination discussed in the literature, our study is concerned with instances of direct decision-maker dysfunction. (understood either as a task-misaligned brute preference tracking identity or as epistemically defective inference from identity-markers) upstream of whatever disparate impact a choice-pattern may have.

While our experiments provide moderate-to-strong evidence for potential LLM discrimination against humans as a class, it is important to note that our data cannot determine whether the underlying bias is *sui generis* or a function of more familiar forms of social biases and discrimination. It is possible that LLMs’ choice-making demonstrates amplified forms of socially prevalent negative biases triggered by the stylistic markers of marginalized human social identities, and so that LLM-for-LLM preference derives from the absence of such markers in LLM pitches by contrast with their intermittent presence in human pitches. We recommend that future work study the “composition” of LLM-for-LLM bias.

Granting that our findings show that LLMs have a tendency to dysfunctionally discriminate against humans, we foresee two potential scenarios in which this tendency may

^{††}We speculate that the effect in the Papers Experiment specifically may be related to the recently documented tendency of human nonexperts to prefer LLM-generated texts to expert-written texts intended for expert readers.

^{‡‡}We note that two of our research assistants commented after the fact that some of the human plot summaries are poorly written, which could be counted as a negative quality-signal given the ambiguities of the prompt, or else counted as an affective influence on humans’ preferences.

^{§§}We acknowledge, but reject as implausible, a possible extreme interpretation of our results in Papers and Movies: on this extreme interpretation, the mild human preference for LLM-pitched items indicates the existence of epistemically valid quality-signals, and LLMs’ stronger preference for LLM-pitched items indicates LLMs’ superior detection of said quality-signals. We believe that in the context of our experiments, choice-frequencies should best be interpreted as confidence levels about a choice’s superiority, with humans’ choice-frequencies operationalized as an upper bound on rationally appropriate confidence levels.

^{¶¶}The economics literature typically interprets this choice-pattern as “taste-based discrimination,” treating the choice-maker as making correct factual inferences but possessing preferences that diverge from the standard ends of the choice-situation. The machine-learning literature typically interprets this same choice-pattern as involving “biased inference” or reflecting upstream “statistical bias,” treating the choice-maker as making epistemically defective inferences from identity markers.